



CARTESIAN PRODUCT AND RELATIONS

CMSC 56 – DISCRETE MATHEMATICS IN
COMPUTER SCIENCE I

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Major Reference Books:

Kenneth H. Rosen (2012). Discrete Mathematics and Its Applications. 7th Edition. The McGraw-Hill Companies, Inc.

K.H. Rosen, J.G. Michaels, J.L. Gross, J.W. Grossman, D.R. Shier (2000). Handbook of Discrete and Combinatorial Mathematics, CRC Press LLC



At the end of this lecture, the students should be able to:

- Define relations on sets
- Identify properties of binary relations
- Apply the definitions to determine the type of a given relation.

Ordered n-tuple

The **ordered n – tuple $(a_1, a_2, \dots, a_n)$** is the ordered collection that has a_1 as its first element, a_2 as its second element, ..., and a_n as its n th element.

Ordered n-tuple

In particular, 2-tuples are called **ordered pairs**.

The ordered pairs (a, b) and (c, d) are equal if and only if $a = c$ and $b = d$.

Cartesian Product

Let A and B be sets. The product set or **Cartesian product of A and B** , denoted by

$$A \times B$$

and read “ A cross B ”, is the set of all ordered pairs (a, b) such that $a \in A$ and $b \in B$, that is

$$A \times B = \{(a, b) : a \in A, b \in B\}.$$

Commonly, A^2 is written as $A \times A$.

Cartesian Product

Note that

$$A \times B \neq B \times A$$

and also

$$n(A \times B) = n(A) \cdot n(B)$$

This follows that for any ordered pair (a,b) in $A \times B$, there are $n(A)$ possibilities for A , and also $n(B)$ possibilities for B .

Binary Relation

Definition:

Let A and B be sets. A *binary relation*, or simply, a relation from A to B is a subset of $A \times B$.

Suppose R is a relation from A to B . Then R is the set of ordered pairs where each first element comes from A and each second element comes from B . Notation for this relation includes xRy , where $x \in A$ and $y \in B$.

Relations

The *domain* of the relation R from A to B is the set of all first elements of the ordered pairs which belong to R , and so it is a subset of A ;

and the *range* of R is the set of all the second elements, and so it is a subset of B .

Relations

Example: Let $A = \{j, y, p\}$ and $B = \{b, d, z\}$,
and let $R = \{(j, d), (j, z), (y, b)\}$.

Is R a relation from A to B ?

What is the domain and range of R ?

Is j R -related to d ? to b ?

Relations

Example: Let $A = \{j, y, p\}$ and $B = \{b, d, z\}$, and let $R = \{(j, d), (j, z), (y, b)\}$.

Is R a relation from A to B ?

- yes, since R is a subset of the Cartesian product $A \times B$

What is the domain and range of R ?

- Domain of $R = \{j, y\}$
- Range of $R = B$.

Is j R -related to d ? to b ? – yes and no

Relations on One Set

The following are examples of relations on a single set A :

1. Let A be a set of people and the relation R describe who likes whom; that is, $(x, y) \in R$ if and only if x likes y .
2. Let $A = \mathbb{N}$, and let xRy hold if and only if $x|y$ (read as “ x divides y ” or “ x is a factor of y ” or “ y is divisible by x ”).
3. Let $A = \mathbb{N}$, and let xRy hold if and only if $x \leq y$.

Universal and Empty Relations

Let A be any set. Then $A \times A$ and $\emptyset$ are subsets of $A \times A$, and hence, are relations on A . They are called the universal and empty relations, respectively.

Thus, for any relation R on A , we have:

$$\emptyset \subseteq R \subseteq A \times A.$$

Inverse Relation

Let R be any relation from set A to set B . The inverse of R , denoted by R^{-1} , is the relation from B to A which consists of those ordered pairs which, when reversed, belong to R ; that is,

$$R^{-1} = \{(b, a) : (a, b) \in R\}.$$

Inverse Relation

Example:

Let $R = \{(1, a), (2, b), (4, z)\}$. What is R^{-1} ?

Inverse Relation

Clearly, if R is any relation, then $(R^{-1})^{-1} = R$. The domain of R^{-1} is the range of R and vice-versa.

Moreover, if R is a relation on $A \times A$, i.e. R is a subset of $A \times A$, then R^{-1} is also a relation on A .

Operations on Relations

Relations are subsets of the Cartesian product, hence they are sets. Thus, all set operations can be applied on relations.

Composition of Relations

Let A , B , and C be sets, and let R be a relation from A to B . Furthermore, let S be a relation from B to C . The composition of R and S , denoted by $R \circ S$, is defined as follows:

$$R \circ S = \{(a, c) : \exists b \in B \text{ with } (a, b) \in R \text{ and } (b, c) \in S\}.$$

In other words, $a(R \circ S)c$ whenever there exists $b \in B$ such that aRb and bSc .

Composition of Relations

Example:

Let $A = \{1, 2, 3, 4\}$, $B = \{a, b, c, d\}$ and $C = \{x, y, z\}$. Furthermore, we let:

$$R = \{(1, a), (2, d), (3, a), (3, b), (3, d)\} \text{ and} \\ S = \{(b, x), (b, z), (c, y), (d, z)\}.$$

Find $R \circ S$.

Composition of Relations

Example:

Let $A = \{1, 2, 3, 4\}$, $B = \{a, b, c, d\}$ and $C = \{x, y, z\}$. Furthermore, we let:

$$R = \{(1, a), (2, d), (3, a), (3, b), (3, d)\} \text{ and} \\ S = \{(b, x), (b, z), (c, y), (d, z)\}.$$

Find $R \circ S$.

Answer: $R \circ S = \{(2, z), (3, x), (3, z)\}.$

Properties of Relations

A relation R on a set A is:

1. **reflexive** if xRx for all x in A

(Everyone likes himself/herself.)

Thus, R is not reflexive if there exists an element a in A such that

$$(a, a) \notin R$$

Properties of Relations

A relation R on a set A is:

2. **irreflexive** if there is no element (a, a) in R , for every $a \in A$
(Nobody likes himself/herself)

Properties of Relations

A relation R on a set A is:

3.a **symmetric** if for all $x, y \in A$, xRy implies yRx .

(If x likes y , then y likes x)

Thus, R is not symmetric if there exists $a, b \in A$ such that $(a, b) \in R$ but
$$(b, a) \notin R$$

Properties of Relations

A relation R on a set A is:

3.b asymmetric if for all $(x, y) \in R$,
 $(y, x) \notin R$.

(If x likes y , then y should
never like x)

Properties of Relations

A relation R on a set A is:

4. **antisymmetric** if for all $x, y \in A$, xRy and yRx imply that $x = y$.

(If x likes y and y likes x , then x and y are the same person)

Thus, R is not antisymmetric if there exists $a, b \in A$ such that $(a, b), (b, a) \in R$ but $a \neq b$.

Note: Every asymmetric relation is also antisymmetric. But if the antisymmetric relation contains (a, a) , then it cannot be asymmetric

Properties of Relations

A relation R on a set A is:

5. **transitive** if for all $x, y, z \in A$, xRy and yRz imply that xRz .

(If x likes y and y likes z , then x also likes z)

Thus, R is not transitive if there exists $a, b, c \in A$ such that $(a, b), (b, c) \in R$ but $(a, c) \notin R$.

Summary of Some Properties of Relations

A relation R on a set A is:

1. **reflexive** if xRx for all x in A
2. **irreflexive** if there is no element (a, a) in R , for every $a \in A$
3. **symmetric and asymmetric**
 - a. **symmetric** if for all $x, y \in A$, xRy implies yRx .
 - b. **asymmetric** if for all $(x, y) \in R$, $(y, x) \notin R$.
4. **antisymmetric** if for all $x, y \in A$, xRy and yRx imply that $x = y$.
5. **transitive** if for all $x, y, z \in A$, xRy and yRz imply that xRz .

Example

Consider the five relations on the set $A = \{1,2,3,4\}$.

1. $R_1 = \{(1,1), (1,2), (2,3), (1,3), (4,4)\}$
2. $R_2 = \{(1,1), (1,2), (2,1), (2,2), (3,3), (4,4)\}$
3. $R_3 = \{(1,3), (2,1)\}$
4. $R_4 = \emptyset$
5. $R_5 = A \times A$

Determine whether the relations are reflexive, irreflexive, symmetric, asymmetric, antisymmetric, or transitive.

Example

Solution:

- a) Only R_2 and R_5 are reflexive.
- b) Only R_3 and R_4 are irreflexive.
- c) Only R_1 and R_3 are not symmetric.
- d) Only R_3 and R_4 are asymmetric.
- e) Only R_2 and R_5 are not antisymmetric.
- f) Only R_3 is not transitive.

Remark

The properties of being symmetric and antisymmetric are not the opposite of each other.

Example:

1. $A = \{(1,3), (3,1), (2,3)\}$
2. $B = \{(1,1), (2,2)\}$

Note:

A is not symmetric and not antisymmetric, while B is symmetric and antisymmetric.